

PREDICATE LOGIC

Discrete Mathematics

PREDICATE

- A predicate is a proposition function
 - Takes as input one or more objects
 - Returns a proposition
- A predicate does not have a truth value
- $P(x) = x > 3$
- “x is greater than 3”.

ANOTHER EXAMPLE

- $P(x, y)$ denotes the statement $x + y = 10$
- A predicate with two parameters
 - Can there be more parameters?
- $P(5, 5)$ is a proposition
- $P(6, 4)$ is a proposition

EXAMPLE

1. Let $P(x)$ denote the statement " $x > 3$." What are the truth values of $P(4)$ and $P(2)$?
2. Let $Q(x, y)$ denote the statement " $x = y + 3$." What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?
3. Let $R(x, y, z)$ denote the statement " $x + y = z$." What are the truth values of the propositions $R(1, 2, 3)$ and $R(0, 0, 1)$?

ADVANTAGES OF PREDICATES

- More general than simple propositions
- We can quantify objects

... WAIT, QUANTIFY??

QUANTIFY

- Other than putting specific values in a predicate, we can also declare that the predicate is true for **every** or **some** values in the domain
- **Quantification** expresses the extent to which a predicate is true over a **quantity** of elements.
- Example: For **every** value of x , $x + 1 > x$
- Example: For **some** value of x , $x < 2$
- Symbols that are used to represent this sort of relation is called a quantifier
 - For **every** object – **Universal** quantifier
 - For **some** object – **Existential** quantifier

UNIVERSAL QUANTIFIER

- Let $P(x)$ denote the statement $x + 1 > x$
- For **every** value of x , $x + 1 > x$
- We can write $\forall x P(x)$
- $\forall$ is called the universal quantifier

EXISTENTIAL QUANTIFIER

- Let $Q(x)$ denote the statement $x < 2$
- For **some** value of x , $x < 2$
- We can write $\exists x Q(x)$
- $\exists$ is called the existential quantifier

EXERCISE

- Let $P(x)$ be the statement " $x + 1 > x$." What is the truth value of the quantification $\forall x P(x)$ where the domain consists of all real numbers?

EXERCISE

- Let $Q(x)$ be the statement " $x < 2$." What is the truth value of the quantification $\forall x Q(x)$, where the domain consists of all real numbers?

EXERCISE

- What is the truth value of $\forall x P(x)$, where $P(x)$ is the statement " $x^2 < 10$ " and the domain consists of the positive integers not exceeding 4?

$X = \{1, 2, 3, 4\}$

$P(4)$ is false

EXERCISE

- Let $P(x)$ denote the statement " $x > 3$." What is the truth value of the quantification $\exists x P(x)$, where the domain consists of all real numbers?

EXERCISE

- What is the truth value of $\exists x P(x)$, where $P(x)$ is the statement " $x^2 > 10$ " and the domain consists of the positive integers not exceeding 4?

TABLE I

TABLE 1 Quantifiers.		
<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x P(x)$	$P(x)$ is true for every x .	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x .